Method of Characteristics

(*) $P(x,y,z)\,z_x + Q(x,y,z)\,z_y = R(x,y,z)$

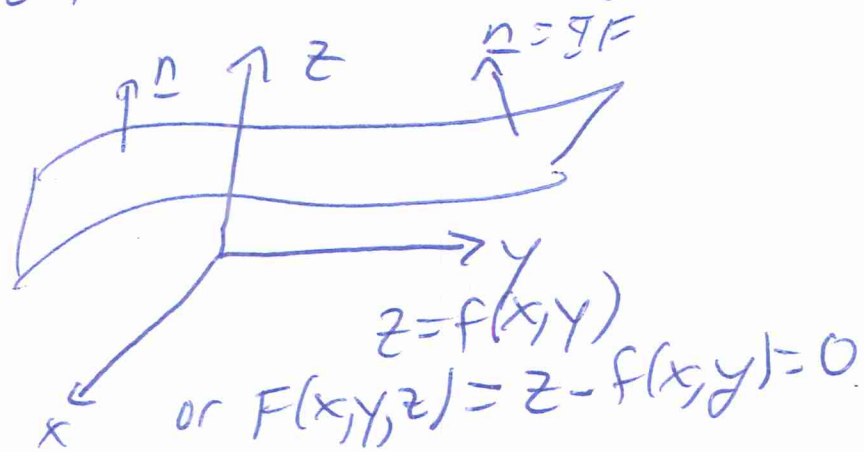
with initial curve $z = \langle x(\tau), y(\tau) \rangle$

added to Friday's notes

soln: $z = f(x,y)$ is a surface.

Recall

(i) Given a ~~sur~~ level surface, $F(x,y,z)=0$, the vector, $\underline{\nabla} F$, is normal to the surface everywhere



(ii) A parametric curve $\langle x(\tau), y(\tau), z(\tau) \rangle$ has a tangent vector $\langle x'(\tau), y'(\tau), z'(\tau) \rangle$

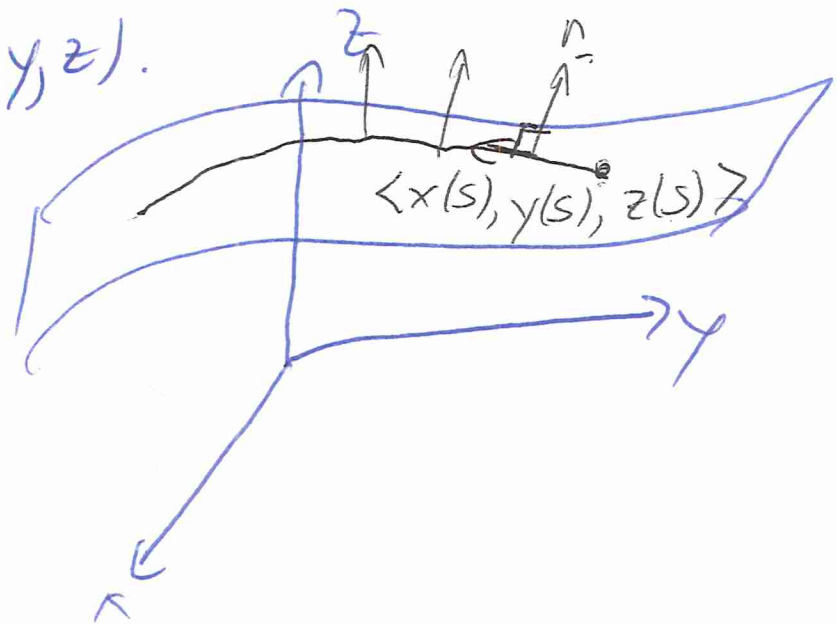
So consider a sol'n, $z = z(x,y)$

It can be ~~can~~ written as a level surface

$$F(x,y,z) = \underbrace{z(x,y)}_{\text{also } f(x,y)} - z = 0$$

Normal to this level surface: $\underline{\nabla} F = \langle z_x, z_y, -1 \rangle$

① By geometry, any curve $\langle x(s), y(s), z(s) \rangle$ that has one pt. on the sol'n surface, and whose tangent, $\langle x'(s), y'(s), z'(s) \rangle$ is everywhere $\perp$ to $\langle z_x, z_y, -1 \rangle = \underline{n}$ remains on the surface $F(x,y,z)$.



② By PDE (#) we have

$$\langle P, Q, R \rangle \perp \langle z_x, z_y, -1 \rangle$$

(dot product is zero)

So:

suppose IC (Initial curve) $z_0 = z(x_0, y_0)$

Then the curve generated by starting at (x_0, y_0, z_0) and which satisfies

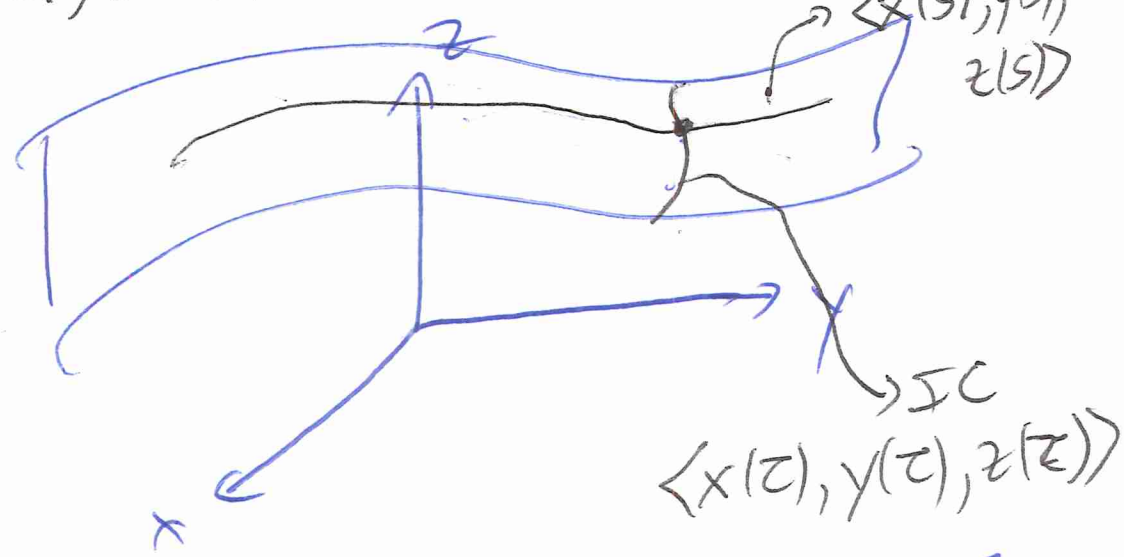
(**) $\langle x'(s), y'(s), z'(s) \rangle = \langle P, Q, R \rangle$ is in the sol'n surface.

$\langle P, Q, R \rangle \perp \langle z_x, z_y, -1 \rangle$

+ by (**) $\langle x'(s), y'(s), z'(s) \rangle \parallel \langle P, Q, R \rangle$

$\Rightarrow \langle x'(s), y'(s), z'(s) \rangle \perp \langle z_x, z_y, -1 \rangle$

+ by ① $\langle x'(s), y'(s), z'(s) \rangle$ is on sol'n surface.



<u>Defn</u> A curve $\langle x(s), y(s), z(s)\rangle$ satisfying

$$\left.\begin{aligned} x'(s) &= P(x(s), y(s), z(s)) \\ y'(s) &= Q(x(s), y(s), z(s)) \\ z'(s) &= R(x(s), y(s), z(s)) \end{aligned}\right\} \text{system of 3 ODE's in 3 unknowns } x, y, z$$

is called a <u>characteristic curve</u> or an <u>integral curve</u>

↳ ref book

$z \perp T$

$\langle x(s), y(s)\rangle$ is known as a <u>characteristic</u> (No 3rd component).

When do we have a sol'n?

A good initial curve $\langle x(t), y(t) \rangle$ is non-characteristic
first 2 components

i.e. Is not // to $\langle P, Q \rangle$. A bad ~~IC~~ I.C. is characteristic.

<u>Thm</u> (Thm 4.2, pg 48, of Z+T)

If $P(x,y,z)$, $Q(x,y,z)$, $R(x,y,z)$ are continuously differentiable ($C^1(\ )$) in a domain (open & connected) in x,y,z-space, and $\langle x_0(\tau), y_0(\tau), z_0(\tau) \rangle$ lies inside the domain and is also continuously differentiable and non-characteristic, i.e.

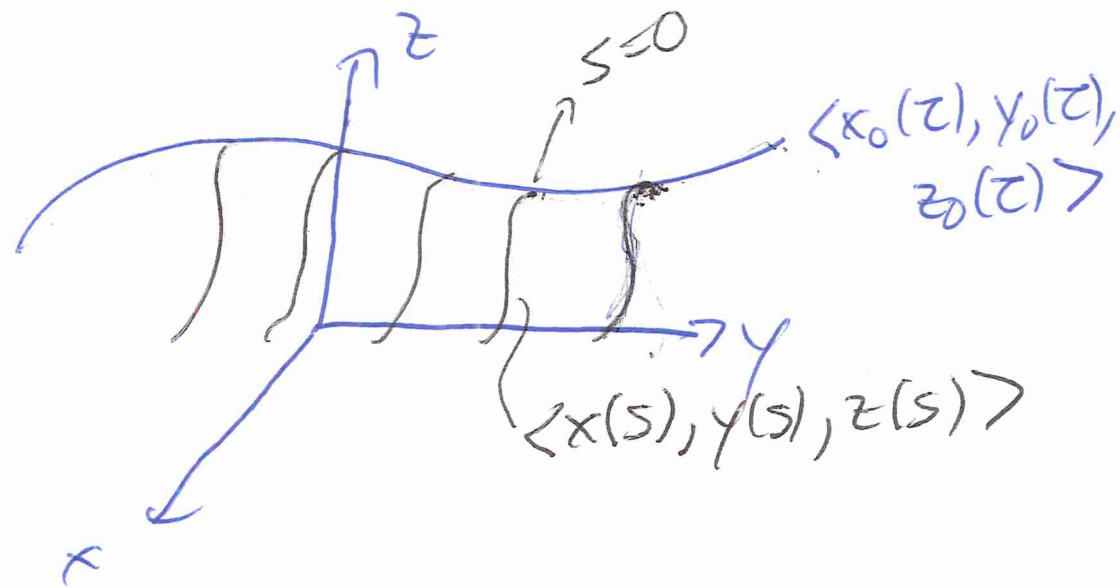
$$\begin{vmatrix} P|_{IC} & Q|_{IC} \\ x_0' & y_0' \end{vmatrix} \neq 0$$

Then there exists a unique continuously differentiable sol'n to

$$P(x,y,z)\, z_x + Q(x,y,z)\, z_y = R(x,y,z)$$

$$z(x_0(\tau), y_0(\tau)) = z_0(\tau)$$

near the I.C.



ex) Farlow p. 207

$$z_x + z_y + 2z = 0 \qquad IC \quad z(x,0) = \sin(x)$$

① Param. The IC if needed $\qquad \langle \tau, 0, \sin(\tau) \rangle$

$\langle x_0(\tau), y_0(\tau), z_0(\tau) \rangle$

② Identify P, Q, R

$P = 1 \quad Q = 1 \quad R = -2z$

③ check conditions for existence + uniqueness

(i) $P, Q, R \in C^1$? Yes

(ii) I.C. charact? $\begin{vmatrix} P|_{IC} & Q|_{IC} \\ x_0' & y_0' \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1 \neq 0$

NO

$\Rightarrow$ we have a unique sol'n :)

④ Find the characteristic curves

$$\frac{dx}{ds} = P \qquad \frac{dy}{ds} = Q \qquad \frac{dz}{ds} = R$$

$\cancel{x_0(s) = x_0}$

$x_0(s=0) = x_0(\tau) \qquad y_0(s=0) = y_0(\tau) \qquad z_0(s=0) = z_0(\tau)$

Solve treating τ as a constant.

$$\frac{dx}{ds} = 1 \qquad \frac{dy}{ds} = 1 \qquad \frac{dz}{ds} = -2z$$

$x_0(s=0) = \tau \qquad y(s=0) = 0 \qquad z(s=0) = \sin(\tau)$

$dx = ds \qquad dy = ds \qquad \frac{dz}{z} = -2\,ds$

$x = s + c_1 \qquad y = s + c_2 \qquad \ln|z| = -2s + c_3$

$s=0 \quad x = \tau \qquad s=0 \quad y=0 \qquad z = c_3 e^{-2s}$

$\tau = 0 + c_1 \Rightarrow c_1 = \tau \qquad \Rightarrow c_2 = 0 \qquad s=0 \quad z = \sin(\tau)$

$\sin(\tau) = c_3 e^0 = c_3$

$\underline{x = s + \tau} \quad (1) \qquad \underline{y = s} \quad (2) \qquad \underline{z = \sin(\tau) e^{-2s}} \quad (3)$

⑤ eliminate s, τ so that sol'n $z = f(x,y)$

$s = y$ ②

① $x = y + \tau \Rightarrow \tau = x - y$

③ $z = \sin(\tau) e^{-2s} = \boxed{\sin(x-y) e^{-2y} = z}$

⑥ check